

PYTHON REINFORCEMENT PROJECT

By : B. Kaviya
Batch : July-2025

Introduction :

DATASET : PHONE USAGE IN INDIA

OBJECTIVE:

- The project focuses on the analyzing and identify key behavioral patterns and relationships in phone usage among Indian users, helping understand how factors like age, gender, and device preference shape digital habits.*

PROBLEM STATEMENT:

- The project aims to analyze how people in India use mobile phones based on their age, gender, and location.*
- It studies patterns in screen time, data usage, and primary phone activities.*



Data Understanding:

Project Goal : Analyze mobile usage behavior across demographics in india.

- The dataset represents simulated **phone usage data for Indian users**. It contains **17,686 rows and 16 columns**, each row representing an individual's phone usage details.
- This dataset contains information about phone usage patterns in India. It includes various **attributes such as age, gender, location, phone brand, operating system, and different usage metrics like screen time, data usage, and more**.





Data Cleaning :

- *The raw dataset contains missing values, incorrect data types, and inconsistent entries, which can lead to inaccurate analysis.*

Steps taken for Cleaning :

- *Handle Data Types*
- *Handle Missing Values*
- *Drop Duplicates*
- *Identify Outliers*

Check for missing values

```
# Checking for missing values:
```

```
print("\nMissing Values:")
print(df.isnull().sum())
```

```
Missing Values:
User ID          0
Age              0
Gender           0
Location         0
Phone Brand      0
OS               0
Screen Time (hrs/day)  0
Data Usage (GB/month)  0
Calls Duration (mins/day)  0
Number of Apps Installed  0
Social Media Time (hrs/day)  0
E-commerce Spend (INR/month)  0
Streaming Time (hrs/day)  0
Gaming Time (hrs/day)  0
Monthly Recharge Cost (INR)  0
Primary Use      0
dtype: int64
```

Check for Duplicate values

```
# Handle missing, incorrect, or invalid data
# Filling missing numeric values with the mean of respective columns

numeric_cols = df.select_dtypes(include=['float64', 'int64']).columns
for col in numeric_cols:
    if df[col].isnull().sum() > 0:
        df[col].fillna(df[col].mean(), inplace=True)
```

```
[27]:
```

```
# Removing duplicates:

print("\nNumber of duplicate rows:", df.duplicated().sum())
df = df.drop_duplicates()
```

```
Number of duplicate rows: 0
```


Handling Missing Values :

APPROACH :

Ensure that missing data not distort analysis by carefully filling or removing gaps.

METHODS USED :

- **Deletion** : Remove rows with completely empty or irrelevant records.
- **Imputation** : Filled missing numeric values with mean/median where appropriate.
- **Standardization** : Replaced missing categorical values with most frequent category(mode)

```
# Checking for missing values:
```

```
print("\nMissing Values:")  
print(df.isnull().sum())
```

```
Missing Values:  
User ID                                0  
Age                                    0  
Gender                                0  
Location                              0  
Phone Brand                           0  
OS                                     0  
Screen Time (hrs/day)                 0  
Data Usage (GB/month)                 0  
Calls Duration (mins/day)             0  
Number of Apps Installed              0  
Social Media Time (hrs/day)           0  
E-commerce Spend (INR/month)          0  
Streaming Time (hrs/day)              0  
Gaming Time (hrs/day)                 0  
Monthly Recharge Cost (INR)           0  
Primary Use                           0  
dtype: int64
```

# Descriptive statistics:				
<pre>print("\nSummary of cleaned dataset:") print(df.describe())</pre>				
Summary of cleaned dataset:				
	Age	Screen Time (hrs/day)	Data Usage (GB/month)	\
count	17686.000000	17686.000000	17686.000000	
mean	37.584247	6.546376	25.411257	
std	13.338252	3.172677	14.122167	
min	15.000000	1.000000	1.000000	
25%	26.000000	3.800000	13.200000	
50%	38.000000	6.600000	25.300000	
75%	49.000000	9.300000	37.600000	
max	60.000000	12.000000	50.000000	
	Calls Duration (mins/day)	Number of Apps Installed		\
count	17686.000000	17686.000000		
mean	151.405846	104.584869		
std	84.923353	55.217097		
min	5.000000	10.000000		
25%	77.325000	57.000000		
50%	150.600000	104.000000		
75%	223.900000	152.000000		
max	300.000000	200.000000		
	Social Media Time (hrs/day)	E-commerce Spend (INR/month)		\
count	17686.000000	17686.000000		
mean	3.252369	5075.707848		
std	1.590223	2871.604841		
min	0.500000	100.000000		
25%	1.900000	2587.500000		
50%	3.200000	5052.000000		
75%	4.600000	7606.000000		
max	6.000000	10000.000000		
	Streaming Time (hrs/day)	Gaming Time (hrs/day)		\
count	17686.000000	17686.000000		
mean	4.250616	2.490874		

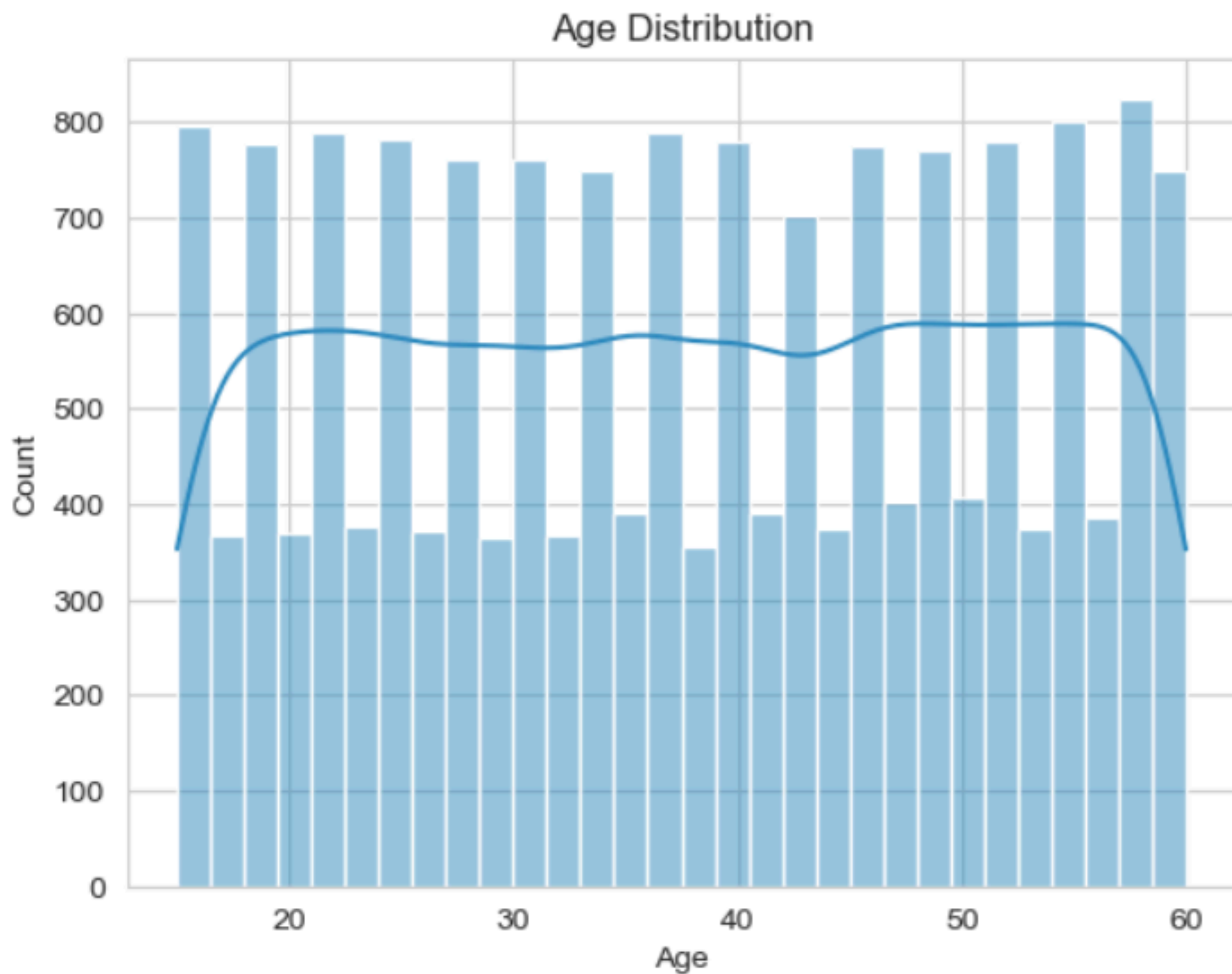
Descriptive Analysis :

- *It is used to summarize and understand the dataset's key characters before deeper analysis.*

Exploratory Data Analysis(EDA)

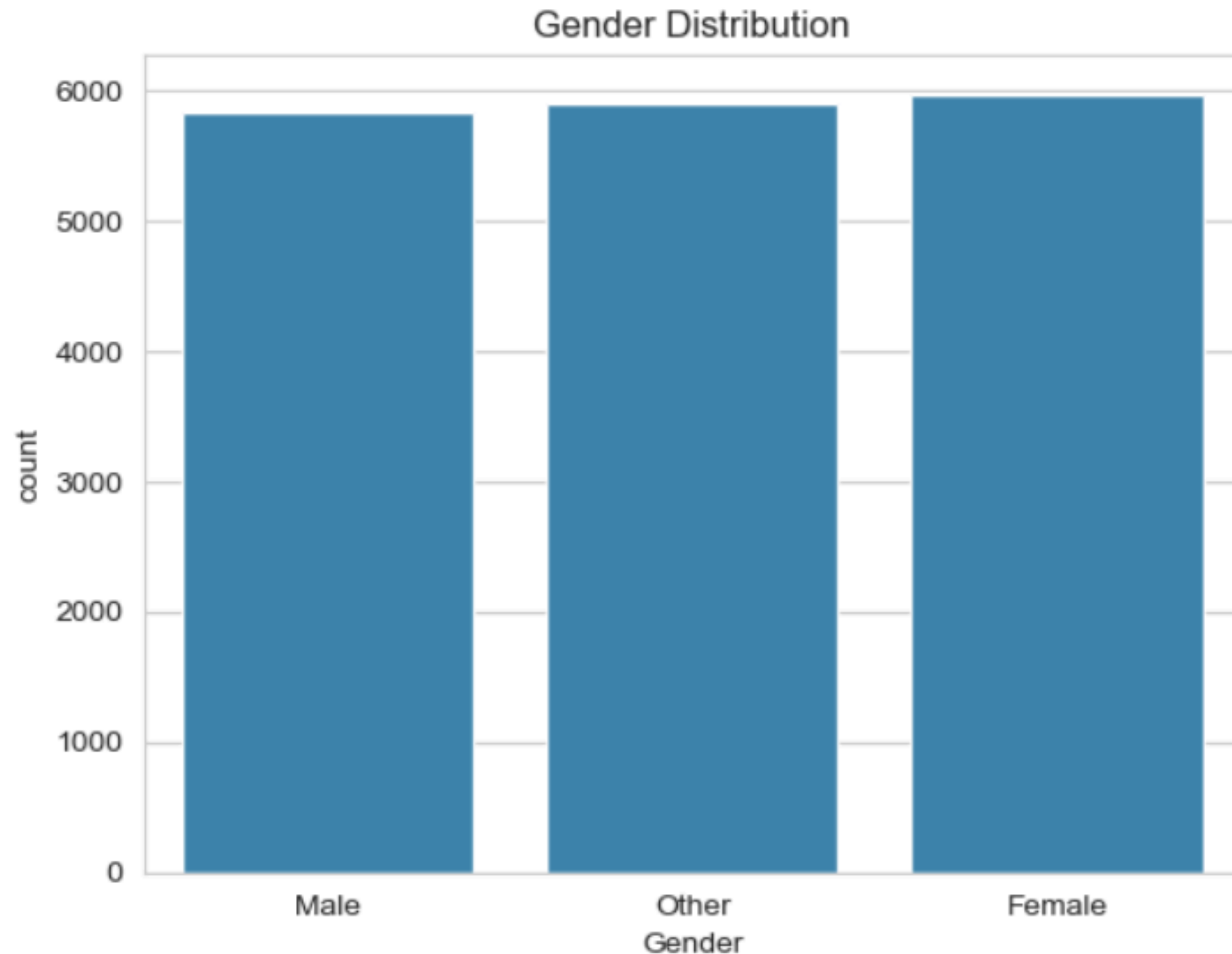
- *EDA is a method of analyzing datasets to summarize their main characteristics, often using visual methods, to gain insights, identify patterns, detect anomalies, and test assumptions without a strict hypothesis.*
- *It is an open, iterative process of examining data to discover hidden information, relationships, and trends, serving as a foundational step for more advanced statistical analysis or modeling*

Age Distribution



- **Many Different Ages:** The data includes people of a wide range of ages, from teenagers to early 60s.
- **Ages are Fairly Evenly Spread:** Most age groups between 20 and 60 have a similar number of people; no one age group heavily dominates.
- **Slightly Fewer Middle-Aged People:** There's a small dip in the number of people in their early to mid-40s compared to other age groups.
- **More Younger and Older Adults:** You might see a bit more people at the younger end (around 20) and a bit more at the older end (around 55-60) of the age range.

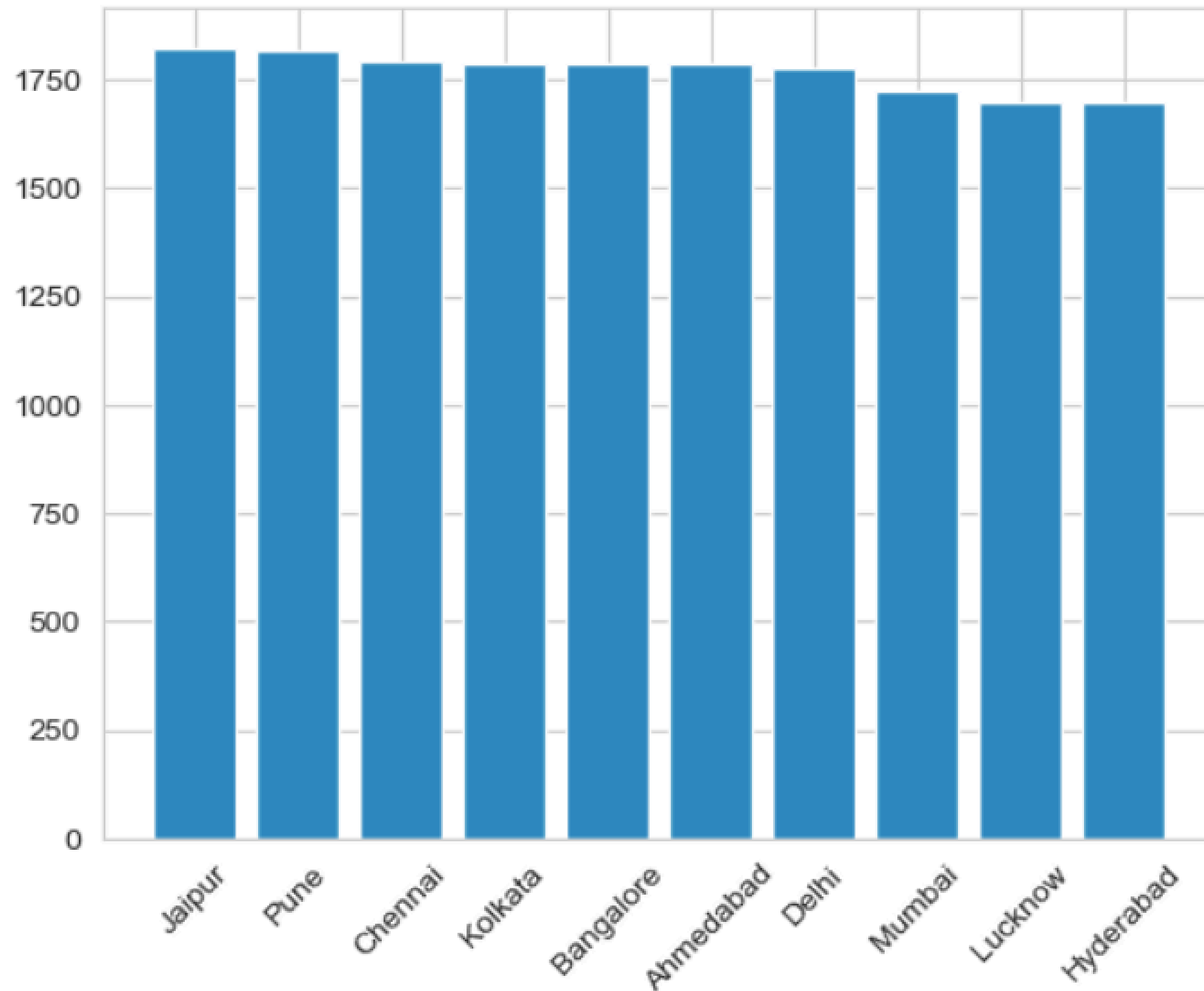
Gender Distribution



- **Equal Representation:** The dataset has a very similar number of people in all three gender categories: Male, Other Gender, and Female.
- **Roughly 5900 Users per Gender:** Each gender category has a count very close to 5900 users.
- **No Gender Dominates:** No single gender group has significantly more or fewer users than the others; they are almost equally represented.

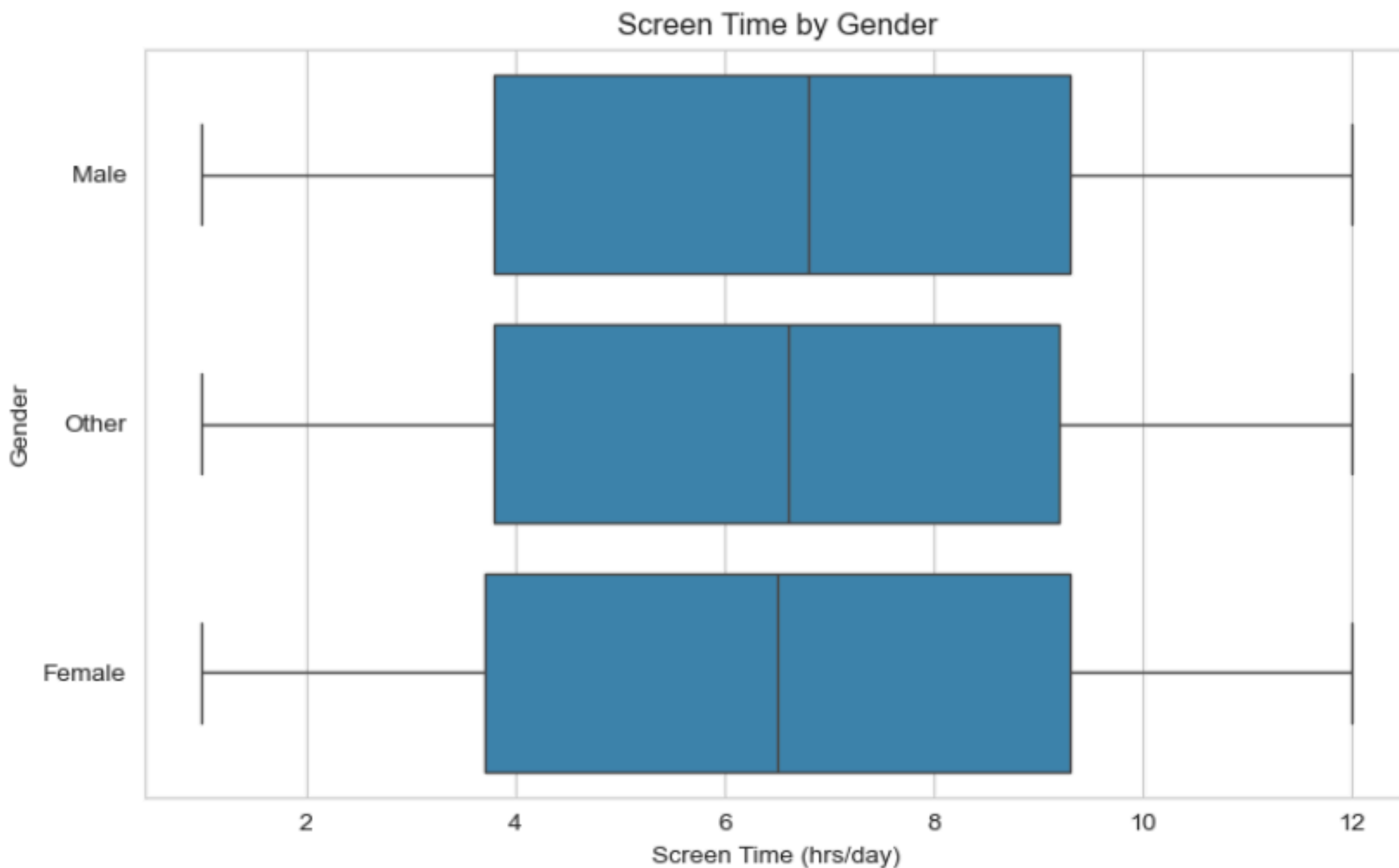
Top 10 Locations

Top 10 Locations by User Count



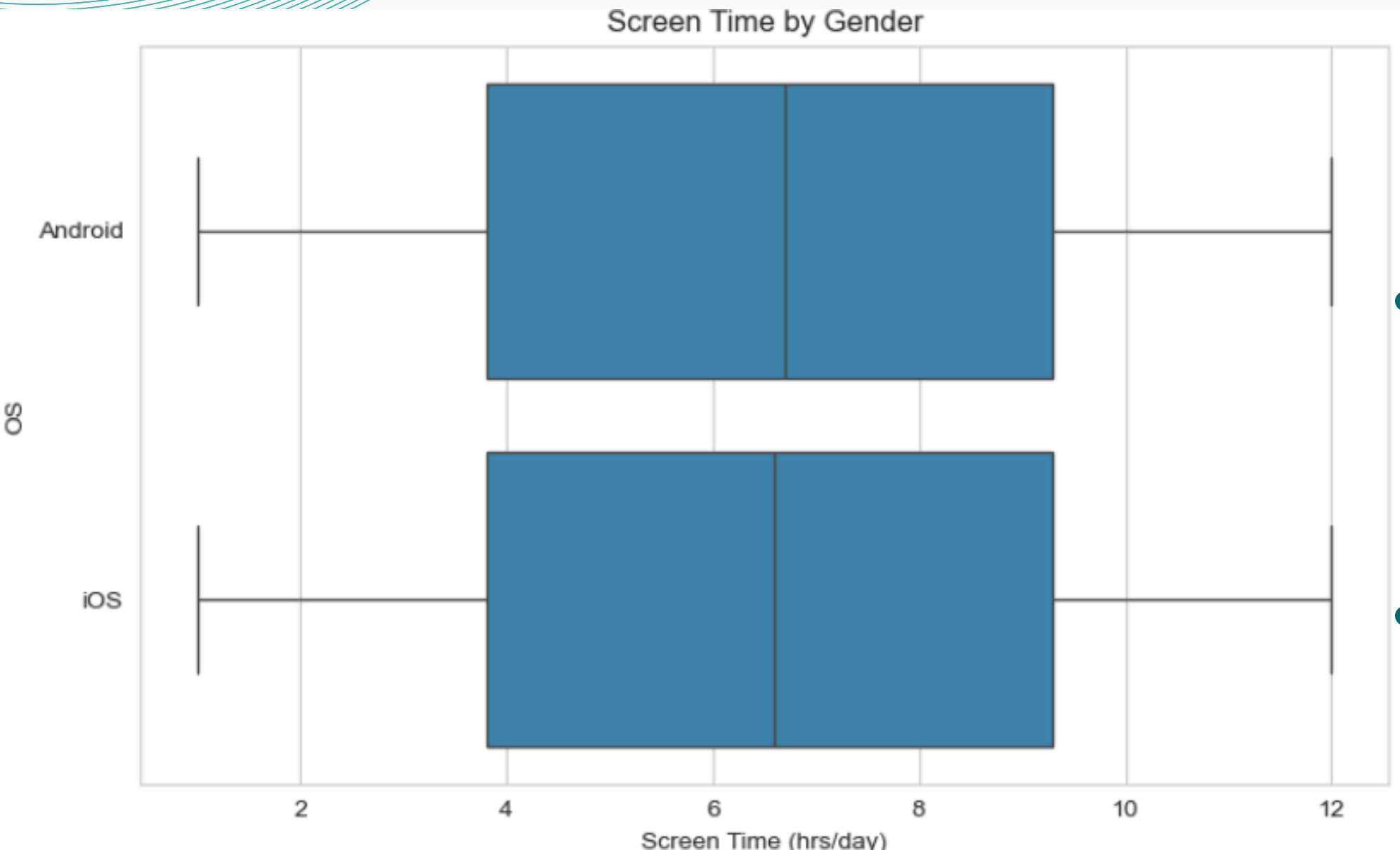
- **Jaipur has the most users:** Jaipur is the location with the highest number of users among the top 10.
- **High User Concentration in Top Cities:** The top few cities (Jaipur, Pune, Chennai, Kolkata) all have very similar, high user counts, closely clustered around 1750 users.
- **Gradual Drop-off Towards the End:** User counts slightly decrease as you move down the list from the top locations to Hyderabad, but the difference between each consecutive city isn't very difference.
- **All Top 10 Locations Have Many Users:** Even the 10th location, Hyderabad, still has a substantial number of users (over 1650), indicating significant user presence across all listed cities.

Screen Time by gender



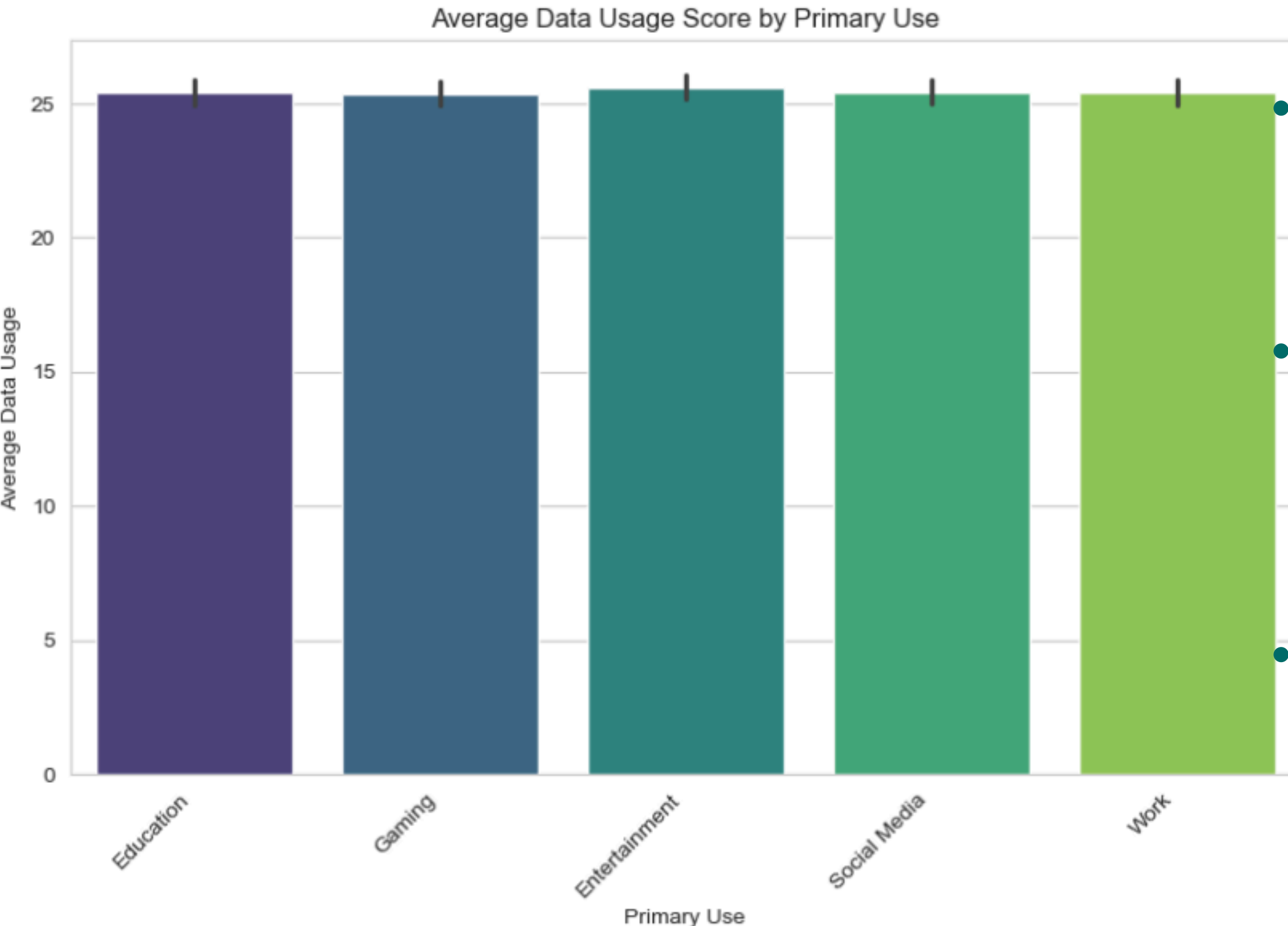
- **Similar Medians:** All genders (Male, Other, Female) have very similar median screen times (around 6-7 hours/day).
- **Consistent Spread:** The spread of the middle 50% of screen time data (Interquartile Range) is similar across all gender categories.
- **Wide Overall Range:** All genders exhibit a wide range of individual screen times, from around 2 hours to over 10 hours daily.

Screen Time by OS



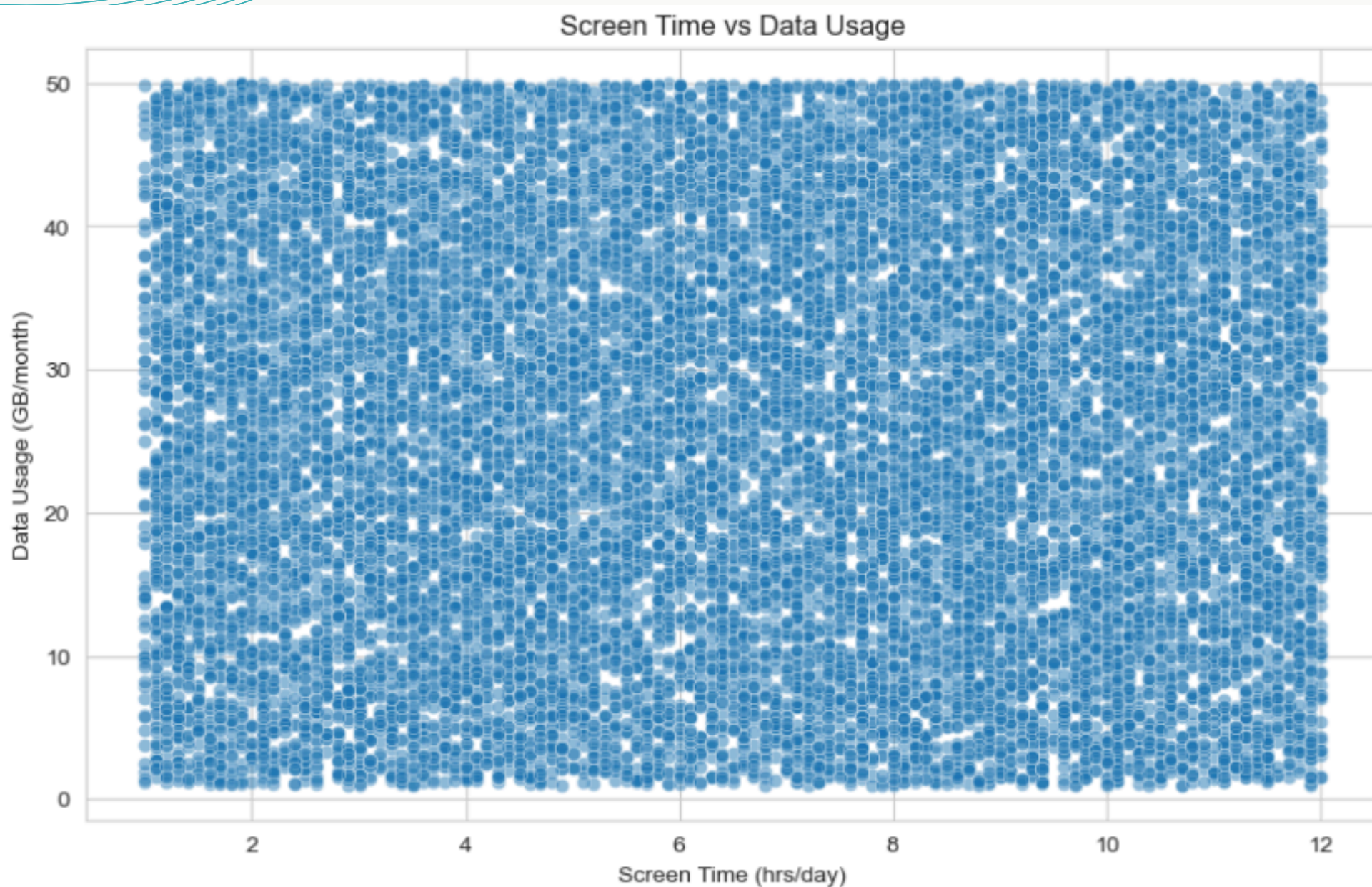
- **Similar Median Usage:** Both Android and iOS users show very similar median daily screen times (the line inside the box), hovering around 6-7 hours.
- **Comparable Spread:** The middle 50% of screen time data (the box itself) is equally spread out for both operating systems, suggesting similar variability.
- **No Significant OS-Based Difference:** The box plots indicate that, generally, the type of operating system (Android vs. iOS) does not strongly correlate with the amount of daily screen time.

Primary Use vs Data Usage



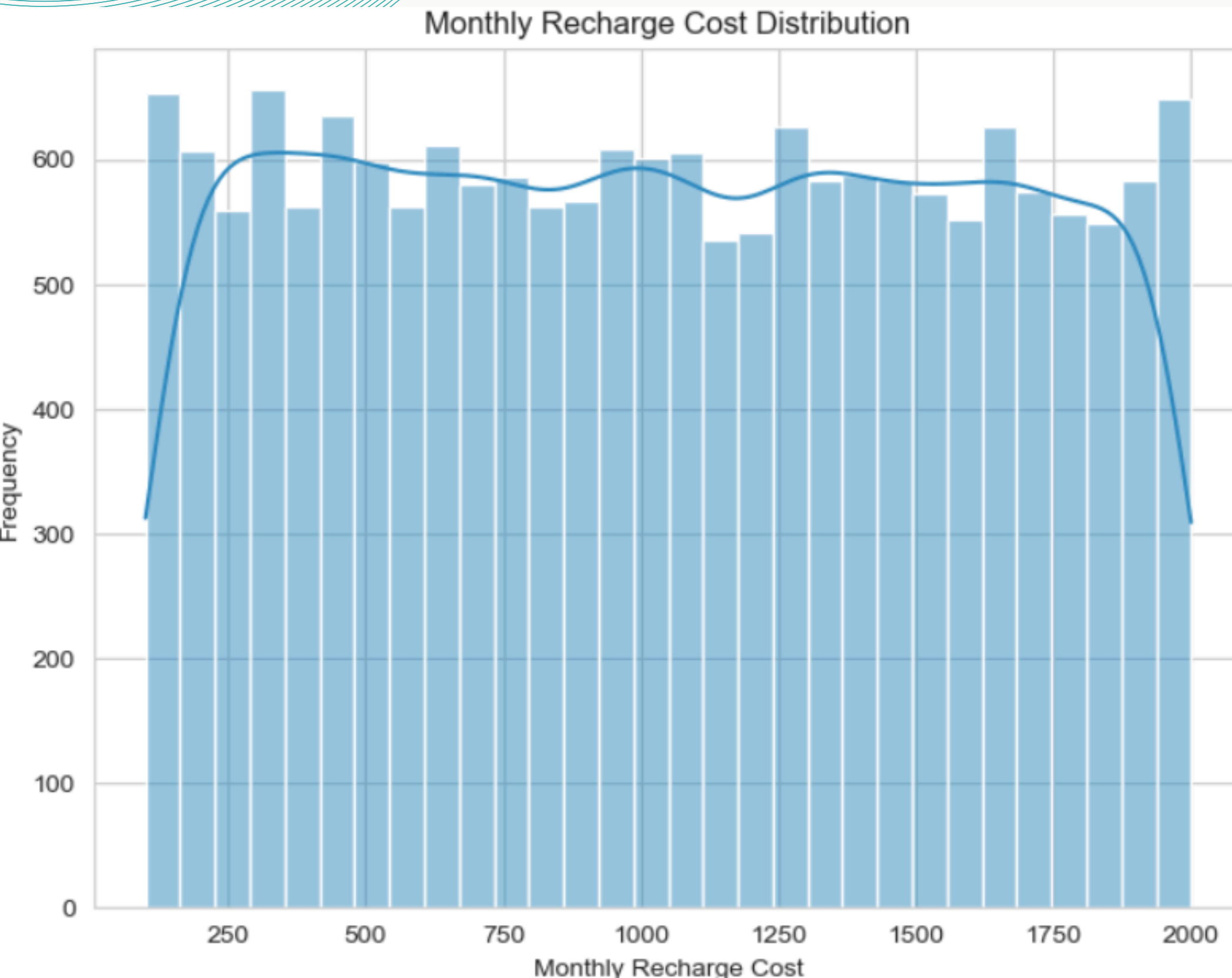
- **Uniform Usage Scores:** The average data usage scores across all primary uses (Education, Gaming, Entertainment, Social Media, Work) are remarkably similar.
- **High and Consistent Scores:** All categories show a high average data usage score, hovering consistently around the 25-point mark.
- **Minimal Variation:** There are no significant differences in average data usage scores between any of the primary use categories, indicating similar data consumption habits regardless of the main activity.
- **Implication:** This suggests that users, regardless of their stated "Primary Use," tend to consume data at a comparable, high level.

Screen Time vs Data Usage



- **No Clear Correlation:** The data points are widely scattered across the graph, indicating no discernible linear relationship between Screen Time (hours/day) and Data Usage (GB/month).
- **Independent Variables:** Changes in screen time do not appear to predictably correspond to changes in data usage, and vice-versa, based on this plot.
- **Uniform Distribution:** Data usage seems largely independent of the amount of time spent on screens, with high and low data usage occurring across all levels of screen time.

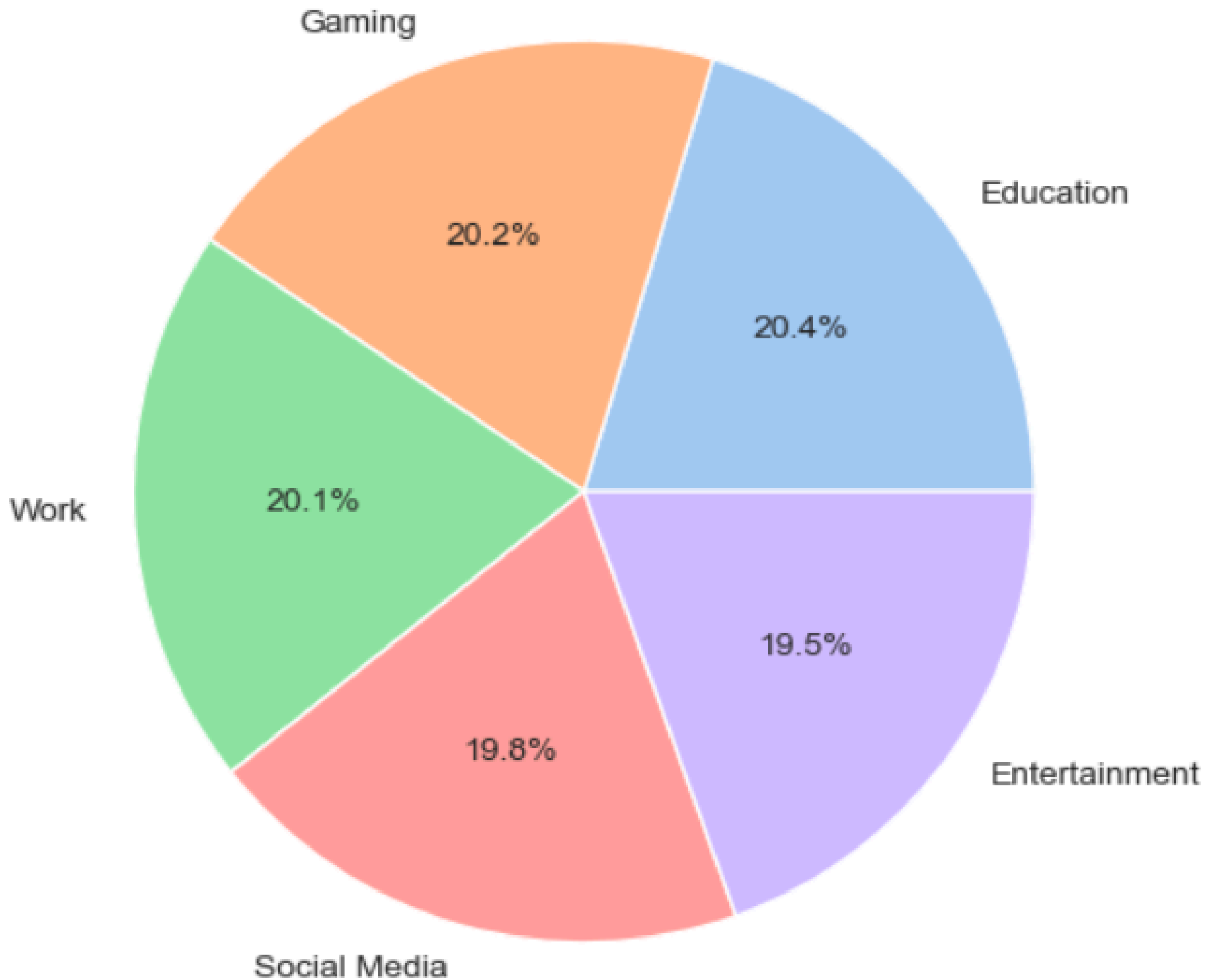
Monthly Recharge Cost Distribution



- **Diverse Spending:** Monthly recharge costs vary widely, from near 0 to over 2000 INR.
- **Multiple Popular Costs:** Several distinct peaks (e.g., around 250, 500, 750, 1250, 2000 - INR) indicate multiple common spending levels.
- **Varied User Habits:** The distribution suggests diverse user spending patterns, not a single dominant cost.
- **No Single Majority:** While there are peaks, no single cost range dominates the entire distribution, implying a lack of a single "average" user in terms of monthly recharge spending.

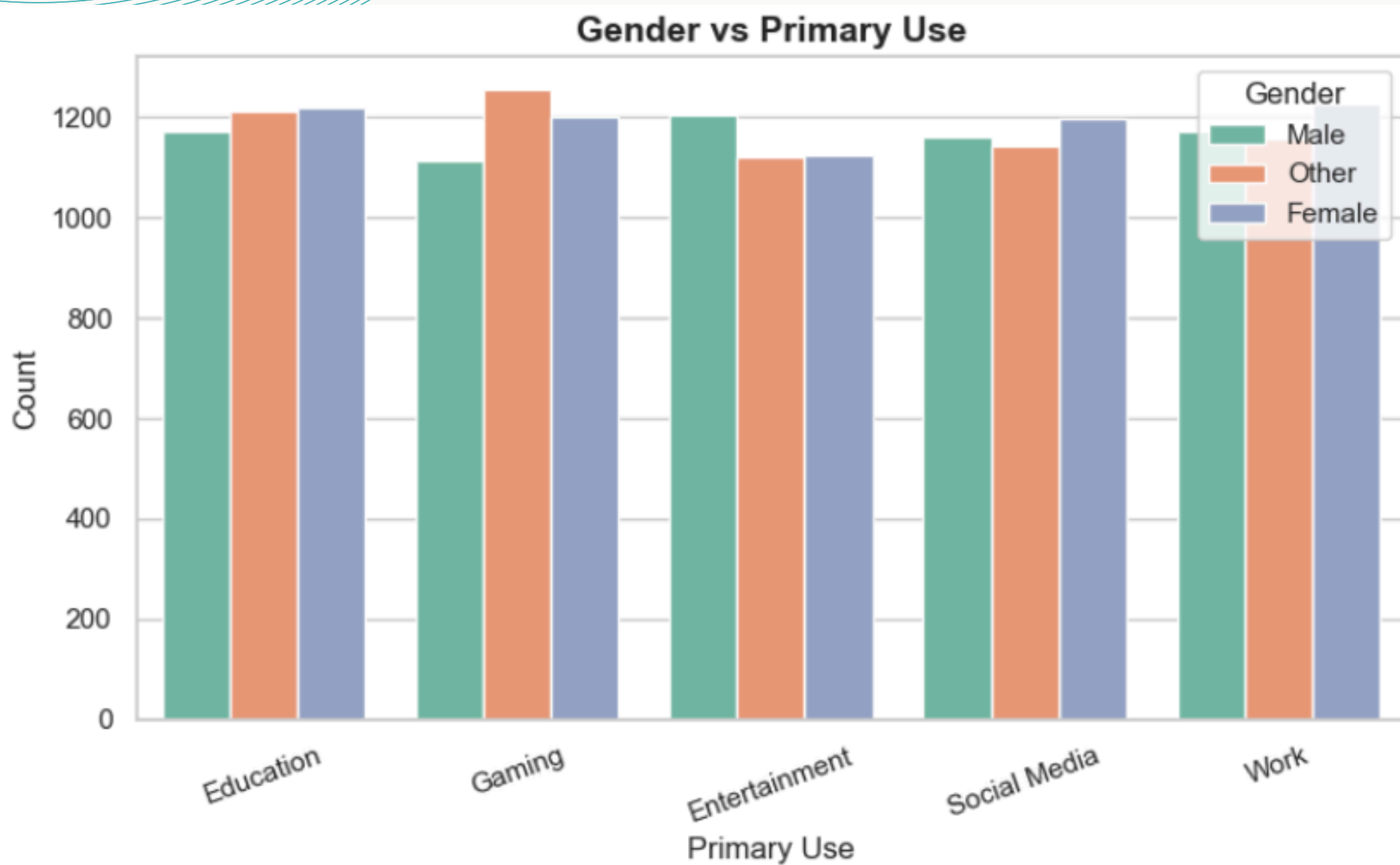
Primary Use

Primary Use Distribution



- **Evenly Distributed Uses:** Primary uses like Education (20.4%), Gaming (20.2%), Work (20.1%), Social Media (19.8%), and Entertainment (19.5%) are remarkably balanced.
- **No Dominant Category:** No single "Primary Use" significantly outweighs the others; usage is spread quite uniformly across these five categories.
- **Near-Equal Shares:** Each category represents roughly one-fifth (around 20%) of the total primary use distribution.

Gender vs Primary Use



- **Gender Distribution:** The data shows a relatively balanced number of users across genders (Male, Female, Other) for each primary use category.
- **Most Popular Use:** "Gaming" has the highest count overall, especially for male users.
- **Female User Trends:** Female users show a higher count in "Education" compared to the other primary use categories.
- **General Consistency:** The counts for all genders are consistently high across all primary use categories, with most categories having counts of 1000 or more.

Statistical Test:

***1] Chi-Square
Test: Gender
vs Primary
Phone Use***

***2] One-Way
ANOVA Test :
Screen Time vs
Gender***

1] Chi-Square Test: Gender vs Primary Phone Use

- **Objective** : To determine whether there is a significant relationship between a user's Gender and their Primary Phone Usage.
- **In other words** — we're testing whether males and females tend to use their phones for different purposes.

Contingency Table:

Primary Use	Education	Entertainment	Gaming	Social Media	Work
Gender					
Female	1219	1124	1203	1197	1226
Male	1171	1207	1115	1160	1172
Other	1211	1120	1258	1144	1159

Chi-Square Test Results:

Chi-Square Statistic = 15.731

Degrees of Freedom = 8

P-Value = 0.0464

Conclusion: $p\text{-value} < 0.05 \rightarrow$ Reject the Null Hypothesis (H_0)

There is a significant relationship between Gender and Primary Use.

Interpretation:

- The $p\text{-value}$ (0.0464) < 0.05 , so we reject the null hypothesis.
- There is a significant relationship between Gender and Primary Use.

2] One-Way ANOVA Test : Screen Time vs Gender

- **Objective** : To determine whether there is a significant relationship between a user's Gender and their Primary Phone Usage Type(e.g., Social Media, Gaming, Communication, Productivity, etc.)
- **In other words** — we're testing whether males and females tend to use their phones for different purposes.

```
One-way ANOVA: Screen Time across Gender
F-statistic: 1.544
p-value: 0.2135
```

```
Conclusion: p-value  $\geq$  0.05  $\rightarrow$  Fail to Reject the Null Hypothesis ( $H_0$ )
There is no significant difference in screen time between genders.
```

Interpretation:

- The p-value (0.2135) > 0.05
 $\rightarrow$ Fail to reject the null hypothesis.
- There is no significant difference in average screen time between genders.

Final Interpretation from EDA and Statistical Tests

- *The dataset provides valuable insights into Indian smartphone usage behavior.*
- *Young adults are the most engaged age group with high screen time and data use.*
- *Gender does not significantly affect total screen time but does influence the type of phone usage.*
- *Android users are more active and have higher engagement than iOS users.*
- *The Chi-Square test confirms behavioral differences across gender categories, while ANOVA shows that overall usage duration remains similar.*
- *These insights can help app developers, telecom companies, and marketers to target user segments effectively and promote digital well-being.*

**Thank You for
Listening**

